Functional Requirements

Priorities follow the supplied feature list:

- Must have = required core capability
- Should have = planned capability with lower priority

Internship status tracking

Feature priority: Must have.

FR-01 [Must have]: The system shall show each student their own current internship status.

FR-02 [Must have]: The system shall provide Academic Advisors and Internship Coordinators with a semester-wide student status overview.

FR-03 [Must have]: The overview shall be filterable by status, including proposal pending, awaiting presentation, approved, missing documents, in internship, returned/dismissed, and internship ended.

FR-04 [Must have]: The system shall show readiness status as ready, not yet complete, or urgent when the start date is near and required conditions are incomplete.

Host-company proposal, approval, replacement, and appeal

Feature priority: Must have / Appeal Should have.

FR-05 [Must have]: The student shall be able to submit the desired host-company information.

FR-06 [Must have]: The student shall be able to schedule a proposal presentation with the Academic Advisor.

FR-07 [Must have]: The Academic Advisor shall record a passed/not-passed result and comments.

FR-08 [Must have]: If a proposal is not approved, the student shall be able to propose another company and schedule another presentation.

FR-09 [Appeal Should have]: The system shall retain previous proposal and decision history.

FR-10 [Appeal Should have]: If a host company cancels acceptance after approval but before the internship starts, the status shall change to “must find a new placement”; the original history shall remain linked.

FR-11 [Appeal Should have]: A student shall be able to request a review of a failed presentation result. The reason and second result shall be stored separately from the original result.

Training eligibility

Feature priority: Must have.

FR-12 [Must have]: The system shall track the required number of preparation-training sessions; the default is five and the value shall be configurable.

FR-13 [Must have]: The student shall be able to upload a certificate as evidence for each training session.

FR-14 [Must have]: The Internship Coordinator shall verify training attendance/certificates and record whether eligibility is complete.

FR-15 [Must have]: The student shall be able to request a review of a failed eligibility result; the request and second result shall be kept as separate history.

Internship documents and request letter

Feature priority: Must have.

FR-16 [Must have]: The system shall track the Request Letter, Acceptance Letter, and Referral/Placement Letter.

FR-17 [Must have]: The system shall show which required documents are present or missing.

FR-18 [Must have]: The system shall reject unsupported file types or files larger than 10 MB immediately and explain the reason.

FR-19 [Must have]: The system shall send reminders for missing documents and stepped follow-up reminders for an outstanding Acceptance Letter; reminder intervals shall be configurable.

Company locations, routes, and instructor assignment

Feature priority: Must have.

FR-20 [Must have]: The student shall be able to pin the host-company location on Google Maps.

FR-21 [Must have]: The system shall display pinned company locations to authorized staff.

FR-22 [Must have]: The system shall support grouping nearby companies to help plan supervision routes; the degree of automation is to be confirmed.

FR-23 [Must have]: The Internship Coordinator shall assign one Supervising Instructor per company/route group according to the defined assignment rule.

FR-24 [Must have]: If a company already has an assigned instructor, the system shall display the existing assignment and require confirmation before replacement.

FR-25 [Must have]: The system shall retain the history of instructor assignment changes, including old/new instructor, person making the change, and time.

Supervision scheduling, scoring, and interview records

Feature priority: Must have.

FR-26 [Must have]: The assigned Supervising Instructor shall schedule each supervision in advance with date, time, and format (online or on-site).

FR-27 [Must have]: The system shall show supervision scheduling status as not yet scheduled, scheduled, or supervision completed.

FR-28 [Must have]: The Supervising Instructor shall record student scores using the approved scoring form/rubric.

FR-29 [Must have]: The Supervising Instructor shall record the student interview result.

FR-30 [Must have]: For on-site supervision, the system shall save records as local drafts if connectivity is unavailable and synchronize them when connectivity returns.

FR-31 [Must have]: The system shall show a persistent pending-sync indication until offline data is synchronized successfully.

FR-32 [Must have]: The Supervising Instructor shall be able to record a problem case during supervision.

Existing-system confirmation

Feature priority: Must have.

FR-33 [Must have]: The student shall be able to declare that daily tasks/data have been recorded in the existing core internship system.

FR-34 [Must have]: The system shall update the student's overall status after the declaration is received.

Internship end, evaluation, report, and closure

Feature priority: Must have.

FR-35 [Must have]: After the planned three-month duration is complete, the system shall change the status to internship ended.

FR-36 [Must have]: The system shall be able to generate a thank-you letter to the host company from a template using stored data.

FR-37 [Must have]: When the host-company evaluation is recorded as complete, the system shall notify the student.

FR-38 [Must have]: The student shall be able to submit an internship report.

FR-39 [Must have]: The system shall retain every report version when the report is revised and resubmitted.

FR-40 [Must have]: Authorized staff/Academic Advisor shall review the report, record comments, and set the result to passed or needs revision.

FR-41 [Must have]: The system shall check the completion conditions: planned duration complete, required minimum supervision count complete, host-company evaluation complete, report passed, and leave records verified if any.

FR-42 [Must have]: The system shall display any unmet completion conditions and shall not enable final closure until all required conditions are met.

FR-43 [Must have]: The Internship Coordinator shall manually confirm final status closure; this action shall not be bypassed automatically.

Problem-case tracking

Feature priority: Must have.

FR-44 [Must have]: The Internship Coordinator or Supervising Instructor shall record returned/dismissed cases and students at risk of not passing.

FR-45 [Must have]: A problem-case record shall include date, reason, recorder, and assistance measures where applicable.

FR-46 [Must have]: The system shall show a chronological problem-case history for each student.

FR-47 [Must have]: When a student is returned/dismissed during the internship, the status shall change to must find a new placement and the student may restart the host-company proposal process.

Notifications and reminders

Feature priority: Should have.

FR-48 [Should have]: The system shall notify Academic Advisors when a proposal presentation is approaching.

FR-49 [Should have]: The system shall notify Supervising Instructors when a supervision schedule is approaching.

FR-50 [Should have]: Internship coordinator shall be able to configure reminder lead time. A seven-day default is suggested but is not confirmed.

Organization master and placement quota

Feature priority: Should have.

FR-51 [Should have]: When a student proposes a host company, the system shall search existing organization records first and allow selection of a matching record to reduce duplicates.

Readiness gate

Feature priority: Must have.

FR-52 [Must have]: Before the internship starts, the system shall check that the host-company proposal is approved, all three required documents are complete, training eligibility has passed, and a Supervising Instructor is assigned.

FR-53 [Must have]: The system shall prevent the internship status from changing to in internship while any required readiness condition is incomplete.

FR-54 [Must have]: The Internship Coordinator shall be able to filter the dashboard to show urgent readiness cases.

Leave and evidence records

Feature priority: Should have.

FR-55 [Should have]: The student shall be able to record leave type, date range, number of days, reason, and supporting evidence after handling the leave with the host company.

FR-56 [Should have]: The Supervising Instructor shall have read-only access to leave information and evidence.

FR-57 [Should have]: The student shall be able to upload general internship evidence such as activity photos or supporting documents.

Non-Functional Requirements

The supplied files define several explicit quality requirements and operating rules. Requirements not specified by stakeholders (such as exact response-time targets) are left for confirmation rather than invented.

Security and Access Control

- NFR-01: The system shall require users to log in before accessing protected internship data.
- NFR-02: The system shall restrict pages and actions according to the user's role.
- NFR-03: The system shall protect student records and uploaded documents from unauthorized access.

Data Integrity and Audit History

- NFR-04: Important workflow decisions and changes shall preserve history rather than silently overwrite prior decisions.
- NFR-05: Appeal results, report revisions, and instructor reassignment history shall remain traceable.
- NFR-06: Final internship closure shall require explicit human confirmation by the Internship Coordinator.

File Validation

- NFR-07: Certificate and required-document uploads shall accept PDF, JPG, or PNG files only.
- NFR-08: Each supported upload shall be no larger than 10 MB.
- NFR-09: Invalid uploads shall be rejected immediately with a message explaining the reason.

Offline Operation and Synchronization

- NFR-10: During on-site supervision, the system shall retain a local draft if there is no internet connection.
- NFR-11: A pending local draft shall not be automatically deleted while awaiting synchronization.
- NFR-12: Synchronization shall retry automatically when connectivity returns, and the interface shall show that data is still pending.

Usability and Workflow Clarity

- NFR-13: The system shall display the student's current status and relevant next steps clearly.
- NFR-14: Dashboards shall support filtering by status and highlighting urgent readiness cases.

- NFR-15: Validation messages shall explain what needs to be corrected.

Configurability

- NFR-16: Training-session requirements shall be configurable.
- NFR-17: Notification lead times and supervision minimum counts shall be configurable.

In Scope

- Role-based access and dashboards for students, Academic Advisors, Internship Coordinators, and Supervising Instructors.
- Student status tracking and semester-wide filtering.
- Host-company proposal, presentation scheduling, approval/rejection, replacement, and appeal history.
- Training eligibility and certificate upload/verification.
- Tracking three internship documents and status
- Map location pins, route grouping support, and instructor assignment.
- Scheduling supervision, recording scores/interview results, and saving on-site drafts when offline.
- Student declaration that daily tasks were recorded in the existing core internship system.
- Internship-end actions, thank-you letter generation, evaluation notification, internship report review, and final closure.
- Problem-case tracking, reminders, readiness checks, leave records, and evidence uploads.

Out of Scope / Boundaries

- Direct API integration with the existing core internship system is not included in the current scope; the student confirms that data was entered there.
- The system is not a leave-approval system. Students must handle leave with the host company first, then record it as evidence.
- Host companies do not have their own accounts in the current proposal. Staff are assumed to record host-company evaluation results on their behalf; this requires confirmation.